Entertainment Content Generator

Course Name: Gen AI

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Problem Statement & Objectives

1. Problem Statement

In the entertainment industry, creating engaging screenplay scenes requires significant creativity, time, and domain knowledge. Writers often struggle to quickly generate structured scenes with proper formatting, dialogues, and storytelling elements.

Traditional scriptwriting processes are manual and time-consuming. Additionally, beginners may lack knowledge about professional screenplay formatting and storytelling structure.

Therefore, there is a need for an intelligent system that can automatically generate screenplay scenes based on user input while maintaining proper script structure and context.

The **Entertainment Content Generator** aims to solve this problem by using **Generative AI and Retrieval Augmented Generation (RAG)** to automatically generate screenplay scenes based on topics, genre, mood, and scene length.

2. Project Objectives

The main objectives of this project are:

- To develop a **web-based application** that generates screenplay scenes automatically.
- To integrate **Generative AI models** for creative content generation.
- To use **Retrieval Augmented Generation (RAG)** to improve accuracy and contextual relevance.
- To implement **vector similarity search using FAISS** for retrieving screenplay knowledge.
- To build a **user-friendly interface** for entering scene parameters and displaying generated scripts.
- To design a **scalable architecture using modern web technologies**.

3. Scope of the Project

The scope of this project includes:

- Generating screenplay scenes based on **user-provided topics and parameters**.
- Retrieving relevant screenplay guidelines from a **vector database**.
- Using **AI models to generate structured dialogue and scene descriptions**.
- Providing an **interactive web interface** for users.

The system is designed for:

- Students learning screenplay writing
- Content creators
- Scriptwriters
- AI-based entertainment applications

However, the system focuses only on **single scene generation** and does not currently support full movie script generation or complex story development.

Proposed Solution

1. Key Features

The Entertainment Content Generator provides several key features:

- **AI-powered screenplay generation**
- **Retrieval Augmented Generation (RAG)** for improved context
- **Vector similarity search using FAISS**
- **User-friendly web interface**
- **Customizable scene parameters** such as genre, mood, and scene length
- **Fast response generation using Groq LLM API**
- **Structured screenplay output format**

These features allow users to quickly generate creative screenplay scenes with minimal effort.

2. Overall Architecture / Workflow

The Entertainment Content Generator system follows a **layered architecture** that integrates a web-based interface, backend processing services, vector retrieval mechanisms, and a Large Language Model for content generation. The architecture is designed to ensure **modularity, scalability, and efficient AI-driven content generation**.

The system uses **Retrieval-Augmented Generation (RAG)** to enhance the quality of generated screenplay scenes. In this approach, relevant contextual information is retrieved from a vector database before generating the final output using the language model.

- **Application Architecture**

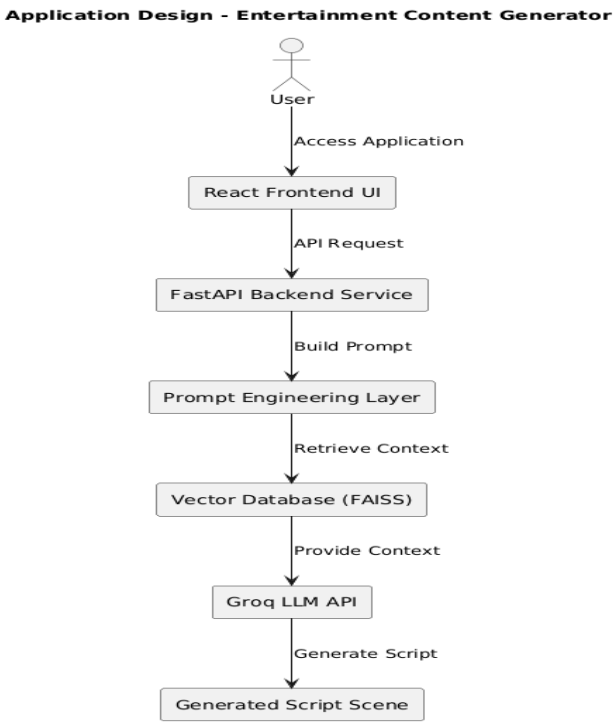


Figure 1: Application Architecture Diagram

This diagram illustrates the main architectural components of the Entertainment Content Generator system. The user interacts with the React frontend interface, which sends requests to the FastAPI backend service. The backend coordinates with the prompt engineering module and the FAISS vector database to retrieve relevant context. The structured prompt is then sent to the Groq LLM API to generate the screenplay scene, which is returned to the user interface.

- **System Workflow**

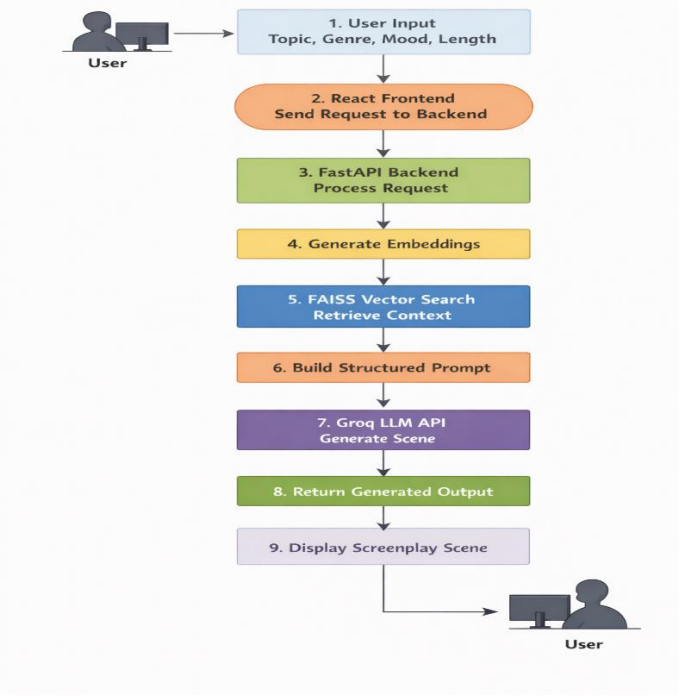


Figure 2: System Workflow Diagram

This diagram represents the complete workflow of the system. The user enters input parameters including topic, genre, mood, and scene length. The React frontend sends the request to the FastAPI backend, where embeddings are generated and similarity search is performed in the FAISS vector database. The retrieved context is used to build a structured prompt, which is then sent to the Groq LLM API. The language model generates the screenplay scene and the final output is displayed to the end user.

- **Information Flow**

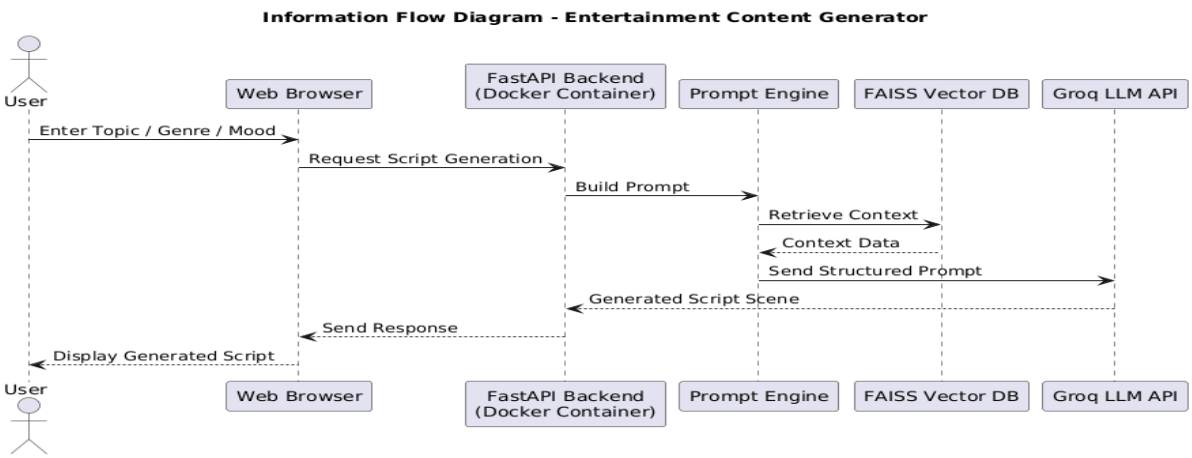


Figure 3: Information Flow Diagram

The information flow diagram shows how data moves between the system components. User input is received through the web browser and forwarded to the FastAPI backend. The backend constructs the prompt using the prompt engine and retrieves relevant contextual information from the FAISS vector database. The structured prompt is then sent to the Groq LLM API, which generates the screenplay scene. The generated result is returned to the frontend and displayed to the user.

- **Component Diagram**

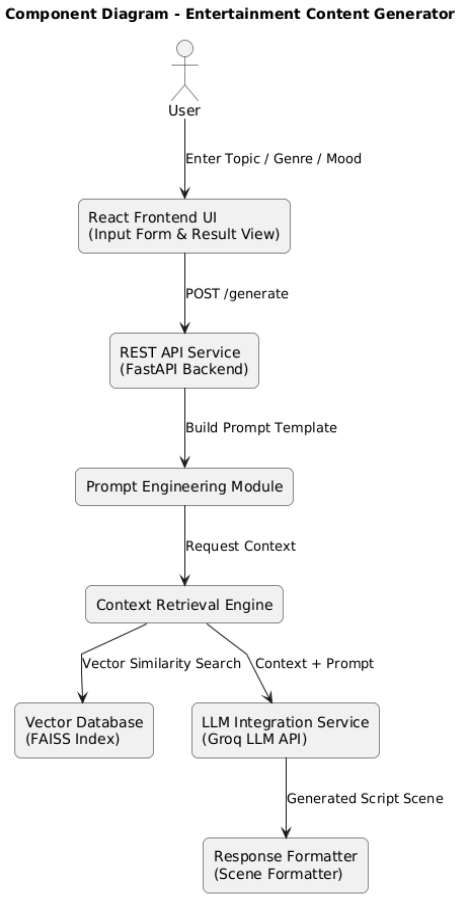


Figure 4: Component Diagram

The component diagram illustrates the major modules of the system and their interactions. The React frontend collects user input and sends API requests to the FastAPI backend. The backend interacts with the prompt engineering module and context retrieval engine. The context retrieval engine performs vector similarity search in the FAISS database and communicates with the Groq LLM API to generate the screenplay scene. The response formatter prepares the final output before it is returned to the user interface.

3. Tools & Technologies Used

Technology	Purpose
React	Frontend user interface
FastAPI	Backend API development
FAISS	Vector similarity search
Groq API	Large Language Model inference
Sentence Transformers	Text embedding generation
Python	Backend programming
HTML/CSS	UI design

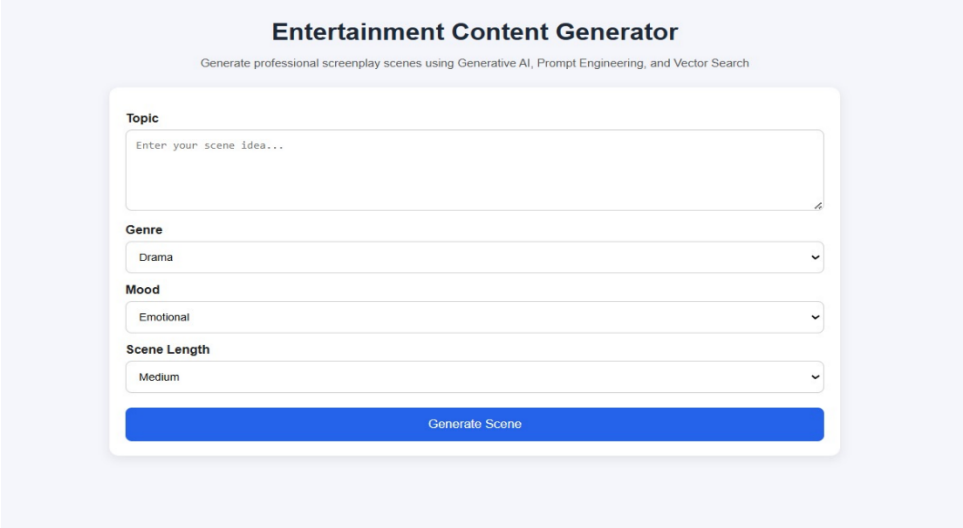
Results & Output

1. Screenshots / Outputs

The system successfully generates screenplay scenes based on user input parameters such as topic, genre, mood, and scene length. The generated output follows a structured screenplay format that includes scene headings, character names, dialogues, and action descriptions.

The following screenshots illustrate the working interface and output of the Entertainment Content Generator system.

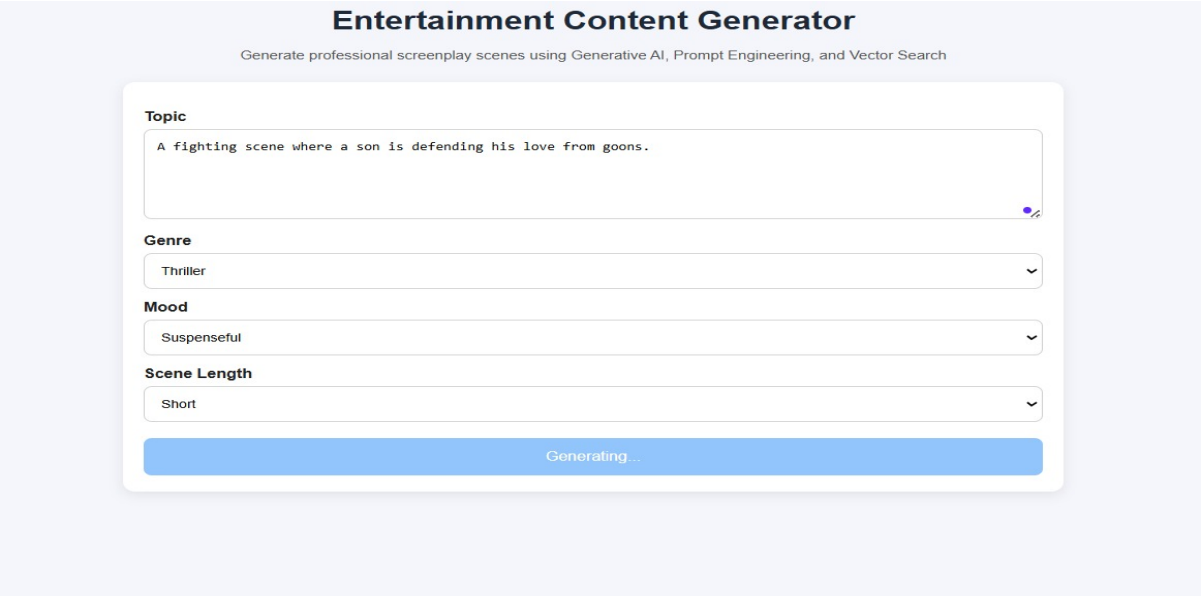
- **User Input Interface**



Caption:

Figure 1: User Input Interface - This screenshot shows the main user interface of the Entertainment Content Generator application. Users can enter the topic and select parameters such as genre, mood, and scene length before generating the screenplay scene.

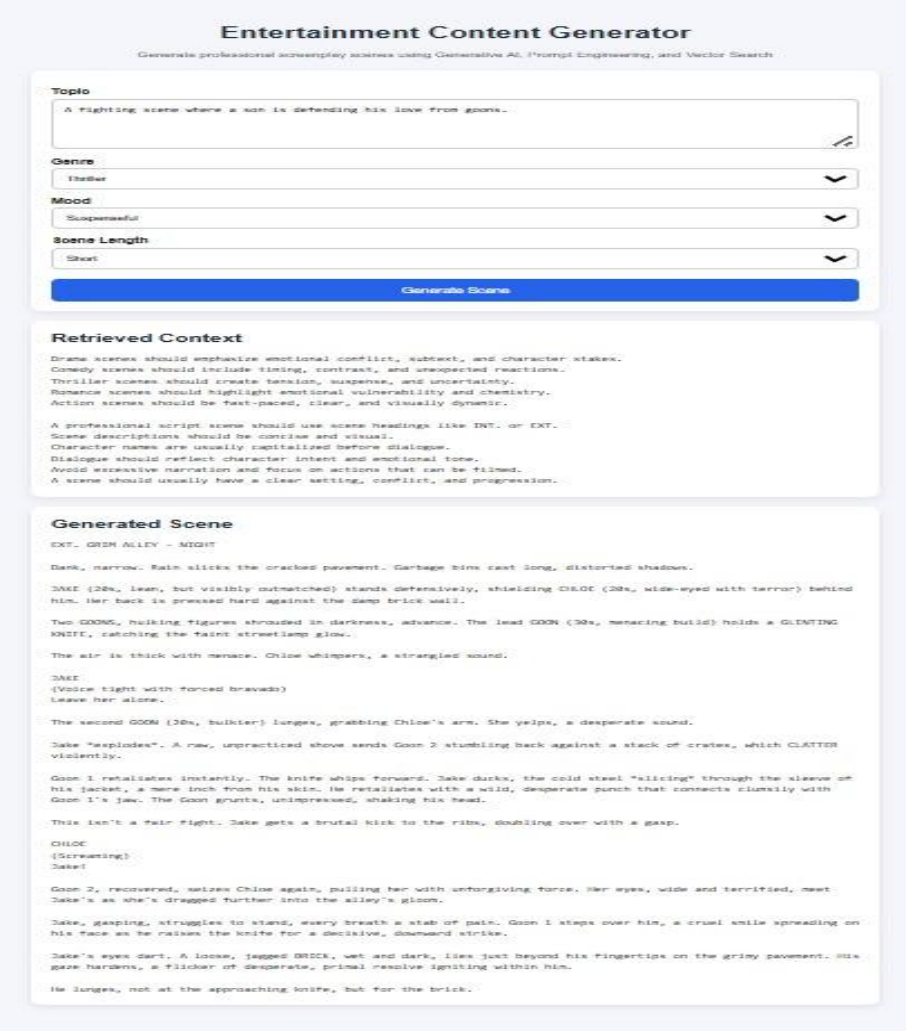
- **User Input with Scene Parameters**

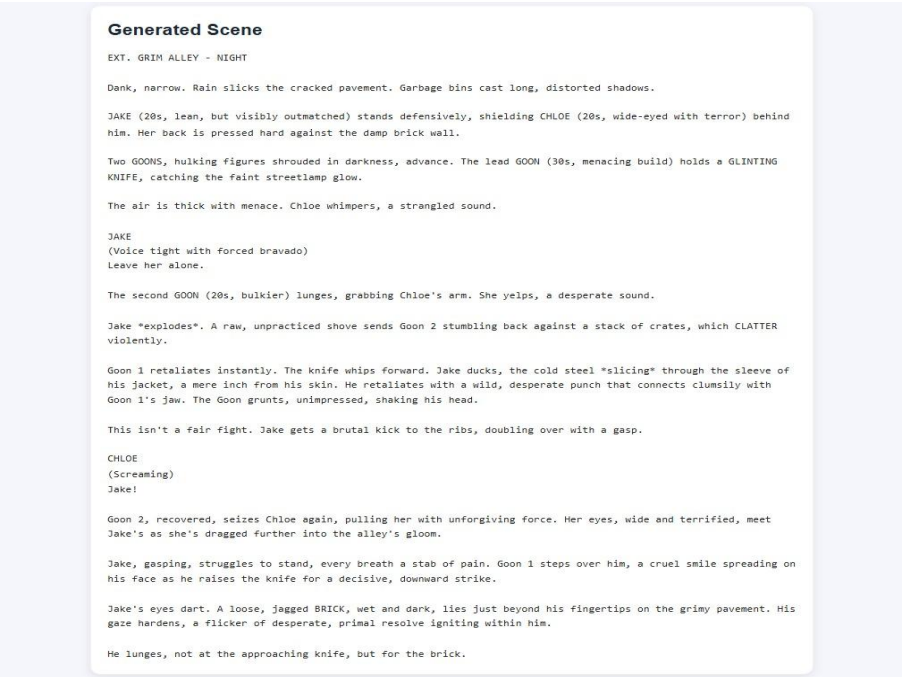


Caption:

Figure 2: Scene Parameter Input - This screenshot illustrates how users provide input details including topic, genre, mood, and scene length. These parameters help the system generate a customized screenplay scene.

- **Retrieved Context and Generated Scene**





Caption:

Figure 3: Retrieved Context and AI Generated Output - This screenshot shows both the retrieved contextual information from the FAISS vector database and the final generated screenplay scene. The retrieved context helps the AI model generate more accurate and structured screenplay content.

2. Reports / Dashboards / Models

The project mainly focuses on **AI-based content generation rather than traditional analytical dashboards or reporting systems**. Instead of visual dashboards, the system relies on a **Generative AI pipeline** that integrates retrieval mechanisms and language models to produce structured screenplay content.

The system includes the following core components:

- **Vector Database (FAISS)** – Stores embeddings of screenplay knowledge and enables efficient similarity search to retrieve relevant contextual information.
- **Prompt Engineering Module** – Constructs structured prompts by combining user input with retrieved contextual information.
- **LLM-based Generation Pipeline** – Uses the Groq LLM API to generate screenplay scenes based on the constructed prompt.

These components work together to form a **Retrieval-Augmented Generation (RAG) pipeline**, which improves the quality, relevance, and structure of the generated screenplay scenes.

3. Key Outcomes

The key outcomes of the project include:

- Successful implementation of a **Generative AI-based screenplay generation system**.
- Integration of **Retrieval-Augmented Generation (RAG)** architecture to improve content quality.
- Efficient **vector similarity search using the FAISS vector database**.

- Fast AI inference using the **Groq Large Language Model API**.
- Development of an **interactive web-based interface using React and FastAPI** for user-friendly content generation.

The project demonstrates how **Generative AI can assist creative industries** by automating parts of the screenplay writing process and generating structured story scenes based on user input.

Conclusion

The **Entertainment Content Generator** project demonstrates the potential of **Generative AI in creative content generation**.

By combining **Retrieval Augmented Generation, vector similarity search, and large language models**, the system can generate structured screenplay scenes based on user input.

The project highlights how modern AI technologies can improve productivity and creativity in entertainment and media industries.

The developed system provides a **scalable and modular architecture** that can be extended with additional AI capabilities in the future.

Future Scope & Enhancements

The system can be further enhanced with additional features such as:

- **Multi-scene story generation**
- **Character generation and development**
- **Story summarization**
- **Genre-specific script templates**
- **Integration with voice or video generation tools**
- **Export scripts in professional screenplay formats (PDF or Final Draft)**

These improvements can make the system more powerful and useful for professional writers and content creators.